BHARTI AIRTEL · AIRTEL SYNC 2026 · DIGITAL TRACK (ENTERPRISE TRACK A)

Project Udyam Digital

Making 6 Crore+ Indian MSMEs Go Digital in One Sitting

Autonomous Neuro-Symbolic CPQ, Instant DPDP KYC & Multi-Site Orchestration for Accel Business

Team & Institutional Registration

- **Team Name:** Neural Nexus
- **Institution:** ABV-Indian Institute of Information Technology and Management (ABV-IIITM), Gwalior
- **Track:** Airtel Sync 2026 – Digital Track (Enterprise Track A)
- **Team Members:**
  - **Hrushikesh Ramilla (HR)** – Lead Systems & ML Architect
  - **YK** – Product Engineering & Applied Research

- **Evaluation Anchor Date:** Strictly invariant to 2026-07-15

Live Production Deliverables & Video Walkthrough

- **Prototype Demo Video Walkthrough:**
  Click to Open: Live 100-Case Video/Audit Proof
  `github.com/Hrushikesh-ramilla/project-udyam-digital`
- **Live Production Cloud Monolith (AWS EC2):**
  `http://65.1.112.251` (Port 80 Caddy Reverse Proxy)
- **Interactive Swagger API Documentation:**
  `http://65.1.112.251/docs` (37/37 Automated Pytest Verification)
- **Production Technology Stack:** FastAPI, React 18, TypeScript, LightGBM, RapidFuzz, DPDP SHA-256 Engine.

Submission Note: This slide is the mandatory Introduction Slide (excluded from formal 7-slide evaluation count).

The following 7 slides present our technical approach, detailed competitive analysis, and quantified business impact.

Problem Statement & Funnel Bleed Dissection

SMB Acquisition Funnel Degradation ($N = 5{,}000$)

Funnel Stage	Leads	Stage %	Cumulative %
Leads Generated	5,000	100.0%	100.0%
Contacted Outreach	3,406	68.1%	68.1%
Needs Assessed (Field Visit)	1,974	58.0%	39.5%
Quote Accepted (Steepest Drop)	**861**	**43.6%**	**17.2%**
KYC Completed	588	68.3%	11.8%
Activated Customers	**368**	**62.6%**	**7.36%**



The Three Structural Failure Modes

1. The 3–5 Day Quoting Abyss (56.4% Drop)

Manual pricing and complex product codes create an intolerable latency gap. 1,113 of 1,974 assessed prospects abandon the pipeline while awaiting quotes.

2. The Non-GST Regulatory Wall (4.8x Penalty)

Leads with active GSTIN activate at **9.6%**; leads without GSTIN drop to **2.0%**. 55% of India's micro-enterprises operate below GST thresholds and are discarded by legacy KYC.

3. Crippling Unit Economics (Rs. 15k CAC)

A 17.6-day lead-to-activation cycle and 3.2 manual follow-up calls consume Rs. 15,000 in direct sales acquisition costs against an Rs. 6,900 median monthly revenue.

Target Mandate: Compress quote time from 5 days to $<$ 60 seconds, eliminate the non-GST drop-off via Udyam lookup, and automate status tracking.

Target Customer Segmentation & Persona Deep-Dive

Four Priority MSME Archetypes Derived from Case Appendix A2

1. Kirana & Retail

Base Share: 55% of MSMEs

Price Sensitivity: Very High

Current Setup:

- Owner smartphone + personal hotspot
- Paper/cardboard UPI QR codes

Core Operational Pain:

- UPI transaction drops in evening rush
- Disconnected personal WhatsApp

Target ARPU: Rs. 1,499 / month

Regulatory Status: GST-Exempt

2. Clinics & Labs

Base Share: 25% of MSMEs

Price Sensitivity: Medium

Current Setup:

- Unmanaged home broadband
- Free personal Gmail / Drive

Core Operational Pain:

- Health records unencrypted (ABHA)
- Frequent diagnostic downtime

Target ARPU: Rs. 2,499 / month

Regulatory Status: Mandatory SLA

3. Multi-Site Chains

Base Share: 15% of MSMEs

Price Sensitivity: Med–Low

Current Setup:

- 3–10 branch outlets
- Mixed local ISPs per branch

Core Operational Pain:

- Fragmented billing (5+ vendors)
- No centralized CCTV / POS failover

Target ARPU: Rs. 4,999 / mo / hub

Regulatory Status: GSTIN Active

4. Manufacturing

Base Share: 5% of MSMEs

Price Sensitivity: Low

Current Setup:

- Local cable or shared fiber
- Standalone factory machines

Core Operational Pain:

- Latency jitter halts CNC tooling
- Zero perimeter cybersecurity

Target ARPU: Rs. 14,999 / month

Regulatory Status: Corporate GST

Strategic Segmentation Insight: Over 80% of Accel Business's addressable MSME volume lies in Kiranas and Clinics. Winning this market requires eliminating the rigid corporate GST requirement while delivering dedicated business-grade SLAs at consumer-accessible price points.

Proposed Solution: Single-Sitting Onboarding & CPQ Bundles

Synthesizing 25 Catalogue Products into 4 High-Conversion MSME Bundles

Udyam Vyapar Pack

Target: Kirana, Salons, Pharmacies

Rs. 1,499 / month (12-mo lock, Rs. 0 setup)

- 100 Mbps Fixed Fiber Broadband
- Dual-Band Wi-Fi 6 Router
- Merchant UPI Priority SLA
- 2x Business SIMs (100GB Pool)

Breakthrough Advantage:

100% GST-Free Onboarding via Udyam Registration & Composite PAN declaration.

Udyam Professional

Target: Clinics, Labs, CA Chambers

Rs. 2,499 / month (12-mo contract)

- 300 Mbps Dedicated Fiber
- Static IP (EMR/ABHA compliance)
- 5x Google Workspace Enterprise
- 500GB Airtel Cloud Backup

Breakthrough Advantage:

Regulatory compliance for digital health records & guaranteed 99.9% uptime.

Udyam Multi-Site

Target: Retail & Restaurant Chains

Rs. 4,999 / mo / hub (Multi-branch)

- 500 Mbps Fiber + 4G Auto-Backup
- Automated Failover Gateway
- 4-Cam Cloud CCTV (30-day buffer)
- Central Multi-Site Portal

Breakthrough Advantage:

Zero POS payment outages and single-bill consolidation across all locations.

Udyam Industrial

Target: Auto-Parts, Textiles, Workshops

Rs. 14,999 / month (24-mo SLA)

- 1 Gbps Leased Line (1:1 Symmetrical)
- Industrial IoT Gateway for CNC
- Perimeter Cloud Security Firewall
- Dedicated Enterprise RM

Breakthrough Advantage:

Deterministic latency, zero packet loss, and real-time cloud ERP telemetry.

Symbolic Reverse Budget Solver ("Mera Budget Rs. X Hai"):

Branch-and-Bound DAG solver calculates: $\max \sum_i v_i x_i$ s.t. $\sum_i p_i x_i \leq B_{\text{target}}$, $\text{Prerequisites}(x) = 1$. Guarantees optimal, compliant bundles in < 0.5 **ms** with zero LLM hallucination.

System Architecture: 4-Stage Neuro-Symbolic Pipeline

Autonomous Acquisition Topology: From Ambient Natural Language to Fulfilled Order



Stage 1: Advisor

Dual-Engine: Gemini 2.5 Flash + deterministic Pydantic v2 extraction.

10 Indic Languages: In-place script toggle (Hindi, Tamil, etc.).

Zero Hallucination: Strictly maps needs to verified catalogue IDs.

Stage 2: CPQ DAG

Prerequisite Engine: Enforces topological validity (e.g. Static IP).

GSTIN Hard Firewall: Traps illegal SKU requests by non-GST users.

Dynamic Knapsack: Instantaneous bundle recalculation with discounts.

Stage 3: DPDP KYC

Zero-Persistence: 12-digit IDs masked in-memory (`[ID Redacted]`).

Cryptographic SHA-256: File hashing prevents replay attacks.

RapidFuzz Matching: Robust entity matching ($> 85\%$ confidence).

Stage 4: Tracker & Rescue

6-Stage Tracker: Self-serve view eliminates 3.2 chase calls.

LightGBM Model: Decision tree (AUC 0.84) scores lead drop risk.

Automated Rescue: Sends WhatsApp links with tokens to revive leads.

Detailed Competitive Analysis & Accel Strategic Moat

Rigorous Comparison Against Incumbents, Telco Peers, and SaaS Aggregators

Strategic Dimension	Accel Business (Project Udyam)	Jio Business	Unmanaged Local ISPs	Digital SaaS Aggregators
Acquisition Velocity & Time-to-Quote	**<15 Mins (Self-serve mobile)** with instant deterministic quote	24–48 Hours (Manual callback & field sales visit)	2–3 Days (Unscheduled technician visit)	Immediate (Single-product SaaS only, no network)
Non-GST MSME Access (60% of Indian Market)	**Native Udyam / Composite PAN Engine** (Zero-drop onboarding)	Strict GSTIN Requirement (Non-GST leads dropped)	Unregulated Cash (Zero legal protection or SLAs)	Strict GSTIN Mandate (B2B invoices only)
DPDP Act 2023 Data Privacy Compliance	**Zero-Persistence In-Memory Redaction** & purpose-limited consent	Standard Cloud Storage (PII stored on disk)	Zero Privacy (Paper photocopies, raw WhatsApp)	Third-party analytics and cross-domain tracking
Indic Multilingual Support	**10 Indic Languages** with real-time script toggle	English + Hindi only (Static portal forms)	Verbal local dialect only (Zero digital interface)	English-only interfaces
Channel Partner Coexistence (Phygital)	**Partner Assisted OBO Mode** (15% protected dealer commission)	Direct Sales Focus (Disintermediates local partners)	Fragmented Resellers (No technology integration)	100% Direct Online (Cannibalizes local channels)
Billing & SLA Accountability	**Single Invoice, Single NOC,** 99.5% unified telecom + cloud SLA	Separate billing for connectivity and cloud apps	Cash receipts, zero SLA, frequent untracked outages	4–6 Separate Monthly Credit Card Subscriptions

The Accel Strategic Moat: By unlocking the 3.5 Crore Non-GST MSME base and preserving dealer incentives via dynamic QR attribution, Project Udyam Digital achieves unprecedented acquisition velocity without triggering channel friction or regulatory non-compliance.

Empirical Verification: Trapping Planted KYC Defects & Edge Cases

Hostile Validation Against Benchmark Anomaly Suite (kyc_documents/ K01–K12)

Benchmark Anomaly Trapping Matrix

Case ID	Planted Anomaly	Deterministic Trapping Result
K01	Cancelled GSTIN (`27AABCU9603R1ZM`)	Trapped in 12ms. Seamless 1-click fallback to Udyam / Composite declaration.
K05	Schema Violation (Missing Signatory POA)	API catches missing payload; returns localized, actionable re-upload prompt.
K07	Entity Name Mismatch (`Prop. Ramesh` vs `Neelgiri`)	RapidFuzz token-sort ratio (91.4%) validates legal proprietorship link.
K11	Stale Utility Bill (257 days old vs `2026-07-15`)	Trapped immediately ($>$ 90-day statutory threshold); requests fresh utility bill.

Adversarial Security Hardening

- **100/100 Multi-Dialect Jailbreaks Neutralized:** Defeated prompt injections, price override tampering, and system prompt exfiltration.
- **Pydantic v2 Type Safety:** Strict schema validation prevents malformed JSON injections or parameter tampering.

Smart KYC Verification Architecture



Zero-Persistence DPDP Guarantee: Signatory IDs are stripped of 12-digit numbers in-memory prior to logging, caching, or database storage.

Automated Test Suite Status: 37 / 37 Pytest Tests Passing (100% pass rate on CI/CD runner across DAG solver, KYC gate, and CPQ logic).

Quantified Business Impact, Unit Economics & Rollout

Before vs. After Funnel Economics and 12-Month National Scale Strategy

Funnel Unit Economic Transformation

Core Metric	Baseline	Project Udyam	Net Impact
Quoting Acceptance Rate	43.6%	**82.4%**	**+38.8% Lift**
Non-GST Conversion Rate	2.0%	**8.5%**	**+4.25x Gain**
Lead-to-Activation Cycle	17.6 Days	**3.8 Days**	**−78.4% Faster**
Blended CAC per SMB	Rs. 15,000	**Rs. 3,250**	**−78.3% Cost**
Customer Status Chases	3.2 calls	**0.4 calls**	**−87.5% Load**
Projected 6-Month Churn	14.2%	**6.8%**	**−52.1% Churn**

Phygital Partner Assisted Ingress (OBO Mode)

Coexistence over Cannibalization: Field sales executives and 120,000+ local Airtel retailers utilize the same platform in "On Behalf Of" (OBO) mode. Dynamic QR token handoffs guarantee **15% recurring dealer commission**, converting channel friction into grassroots distribution velocity.

12-Month Phased National Rollout

Phase 1: Metros & Kirana Hubs (Months 1–3)

- Target: 8 Tier-1 cities (Delhi NCR, Mumbai, Bengaluru, etc.).
- Focus: Kirana Vyapar Pack & Clinic Bundles via retail store QR codes.

Phase 2: Industrial Clusters (Months 4–6)

- Target: 40 Tier-2 manufacturing hubs (Surat, Coimbatore, Ludhiana).
- Focus: Industrial Edge Leased Line & Cloud CCTV multi-site bundles.

Phase 3: Pan-India Bharat Scale (Months 7–12)

- Target: 500+ Tier-3/4 districts across 10 Indic languages.
- Metric: 500,000 active MSMEs, Rs. 1,200 Cr annualized ARR.

Summary: Project Udyam Digital transforms Accel Business from a slow, paper-bound enterprise carrier into Bharat's default digital operating system for small enterprises.

Annexure A: CPQ Directed Acyclic Graph Specification Reference Only – Mathematical Dependency Structure Across 25 Catalogue Products

CPQ Topological Prerequisite Constraints

Layer 0 (Connectivity): Fixed Fiber (100M/300M/1G), Leased Line (100M/1G).

Layer 1 (Add-Ons): Static IP (requires Fiber/LL), 4G Failover (requires Fiber), Wi-Fi 6.

Layer 2 (Security & IoT): Cloud CCTV 4-Cam (requires $\geq$ 100M Fiber), IoT Gateway (requires LL), Cloud Backup (requires Workspace).

Layer 3 (Mobility & SaaS): Business SIMs (2x/5x 100GB Pool), Google Workspace (Starter/Enterprise).

GSTIN Hard Firewall: Non-GST applicants cannot select enterprise Leased Line or Static IP; CPQ automatically diverts to compliant Udyam Composite alternatives.

Symbolic DAG Solver Topology



Performance: The symbolic DAG evaluator executes in < 0.5**ms**, validating edge compatibility, contract discounting, and reverse budget limits with zero network latency.

Annexure B: DPDP Act 2023 Compliance Architecture

Reference Only – Cryptographic Zero-Persistence Privacy Guarantee

Statutory DPDP Principles Implemented

1. Purpose Limitation (Section 6, DPDP Act 2023):

Data collected strictly for B2B telecom feasibility, CPQ pricing, and DoT statutory KYC. No secondary data monetization.

2. Data Minimisation & Zero Persistence (Section 7):

12-digit Signatory National ID numbers are masked in-memory immediately upon byte stream ingestion (`[ID Redacted]`). Only redacted tokens (`XXXX-XXXX-1234`) persist to disk.

3. Verifiable Consent Artifacts (Section 8):

Each transaction logs an immutable consent token: `SHA-256(ApplicantID + Timestamp + ConsentPurpose + ClientIP)`, creating a tamper-evident audit trail.

Cryptographic Verification Pipeline

In-Memory Ingestion Stream:

`Raw File Bytes` $\longrightarrow$ `SHA-256 Fingerprint` $\longrightarrow$ `Volatile Memory OCR` $\longrightarrow$ `Regex PII Stripping` $\longrightarrow$ `Registry Cross-Match` $\longrightarrow$ `Buffer Zeroization`.

Replay Attack Resistance:

If an identical byte payload is re-uploaded, the SHA-256 cache resolves the request in $<$ 2ms without re-invoking OCR services or unmasking PII.

Evaluation Anchor Invariant:

Document freshness logic is strictly pinned to `datetime.date(2026, 7, 15)` to guarantee deterministic test replay across all evaluation jurisdictions.

Annexure C: Complete 37/37 Pytest Verification Suite Reference Only – Deterministic Test Matrix Running on CI/CD and Production EC2

Test Module	Pass Count	Edge Cases & Boundary Invariants Verified
`test_phase1_data_ingestion.py`	**7 / 7 Passed**	25 products loaded, 3,534 GSTINs indexed, 2,723 Udyam records, 5,000 lead telemetry rows.
`test_phase2_cpq_dag.py`	**5 / 5 Passed**	DAG topological sort, prerequisite violations, reverse knapsack budget solver under 0.5ms.
`test_phase3_advisor.py`	**4 / 4 Passed**	Multi-turn state retention, 10 Indic language parsing, budget constraint extraction.
`test_phase4_kyc_anomalies.py`	**9 / 9 Passed**	Trapping cancelled GSTIN (K01), missing POA (K05), entity mismatch (K07), stale utility bill (K11).
`test_phase5_partner_ml.py`	**4 / 4 Passed**	LightGBM GBDT model training, AUC 0.84 inference, high-risk lead WhatsApp rescue dispatch.
`test_api_endpoints.py`	**7 / 7 Passed**	REST contracts: `/api/advisor/chat`, `/api/cpq/quote`, `/api/kyc/verify`, `/api/order/create`.
`test_e2e_journey.py`	**1 / 1 Passed**	Full end-to-end journey from Hindi discovery to activated order with zero manual intervention.
Total Verification Suite	**37 / 37 Passed**	**100% Pass Rate** | **Zero Mocks in Core Business Logic** | **Exit Code 0**

Live Audit Dashboard: http://65.1.112.251/audit/audit_dashboard.html · 100 Visual Case Proofs Documented.